

Unit ID: 679b0f3faaf593b2823f314c

Unit 3 - Creating Your Own Smart Wastebin

PDF Documents

PDF Document 1: 1738215229542-Chapter 6 - Unit 3 - Creating Your Own Smart Wastebin.pdf

This PDF document is included in the unit materials.

Update #1 - 2025-05-16 15:26:12



UNIT 3 CREATING YOUR OWN SMART WASTEBIN

WHAT IS A SMART WASTEBIN?



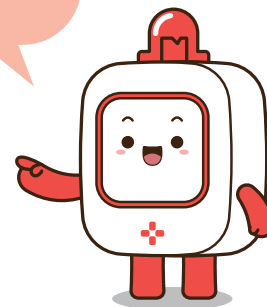
A smart wastebin is a special trash can that opens automatically when it detects someone nearby. This helps keep our hands clean and makes throwing away trash easier. It uses technology like sensors and motors to work without us having to touch it, which is more hygienic and helps prevent the spread of germs.



How It Works in Action:

The smart wastebin uses a ToF (Time of Flight) sensor to detect when someone is close. When you approach the bin, the sensor measures the distance and triggers a motor that automatically opens the lid. This means you don't need to touch the bin to throw away trash. After you step away, the lid will close on its own, keeping everything neat and clean.

By using a smart wastebin, we can make cleaning up easier, more fun, and much more hygienic!



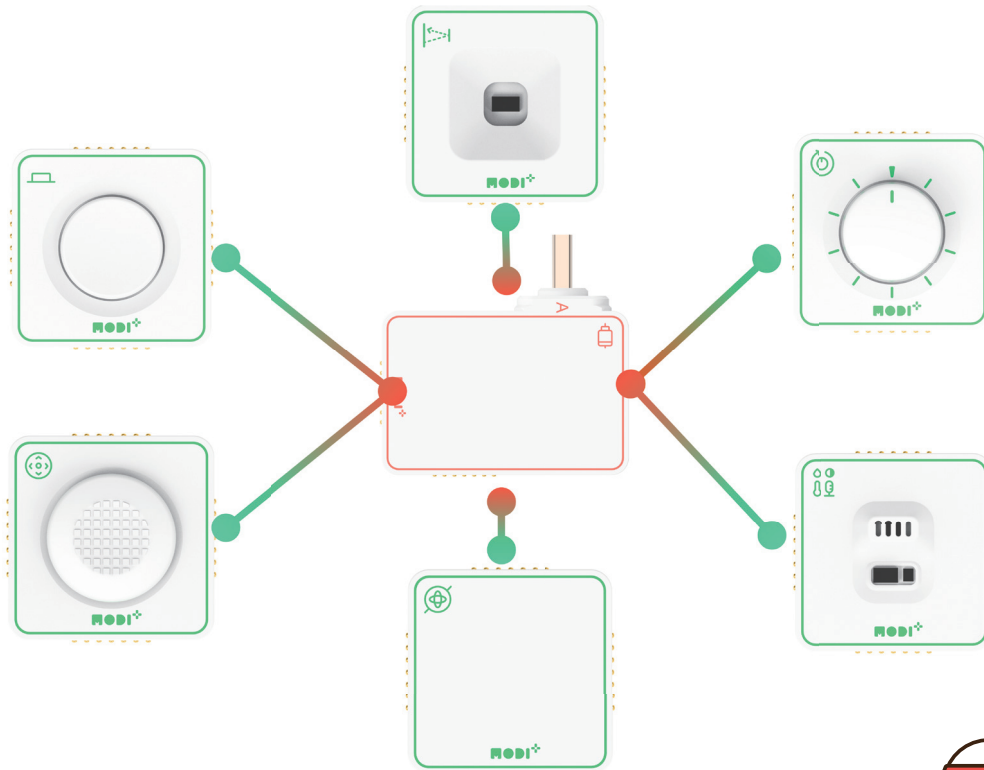
ABOUT MOTOR A MODULE

Motor A features five different blocks in the block area. These blocks allow you to customize your creations by setting the motor's angle, spinning direction, and speed.

INTERACTIONS WITH OTHER MODULES

As an output module, Motor A can interact with all input modules, showing a unique response to each one. Explore these interactions by connecting Motor A to different input modules!

INTERACTION BETWEEN LED AND OTHER MODULES





Activity: Create **Your Own Smart Wastebin**

1 Materials Needed:



Network



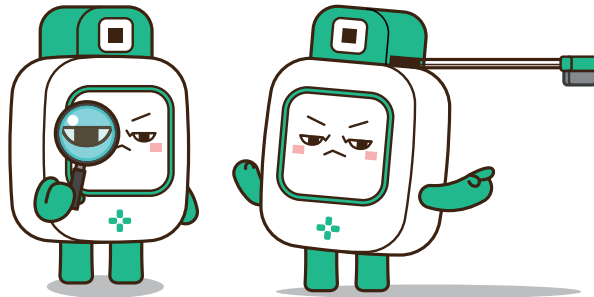
ToF



Motor



I-Horn



2 Code story

Set Motor A's angle to 0



When ToF's distance is less than 10cm

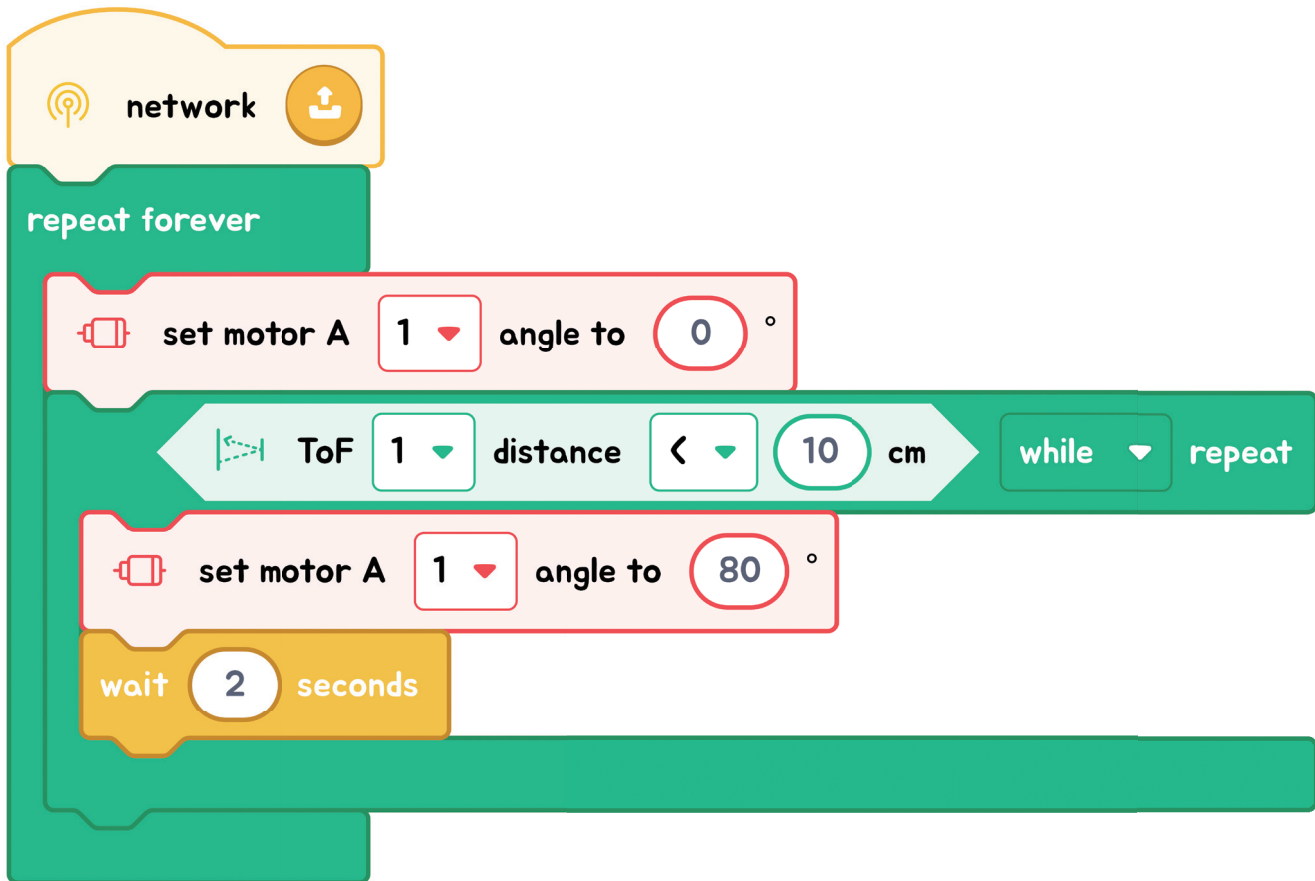


Set Motor A's angle to 80



Wait for 2 second

Code Quest



Checkpoint

Let's add code that notifies with a sound when the wastebin's lid is opened.

